Suggested questions are 3, 4d, 5, 6 and 7.

Qu. 1.

a) Find the equation of the tangent plane to the surface
$$z = x^3 - y^3 + \sin(xy)$$
at the point $(0, -1, 1)$.

b) Let $f(x, y, z) = xyz + x^2y^3$. Find the directional derivative of f at the point $(0, -2, 3)$ in the direction to the point $(4, 1, 2)$.

Qu. 2. For each of the following functions find (i) ∇f and (ii) the directional derivative in the direction of $\mathbf{u}$:

a) $f(x, y) = y \sin x$ where $\mathbf{u} = 2\mathbf{i} - \mathbf{j}$;
b) $f(x, y, z) = xe^{yz}$ at the point $(\pi, 1, -2)$ where $\mathbf{u} = \mathbf{i} - \mathbf{j} + 3\mathbf{k}$;
c) $f(x, y) = x^2y + y^3x$ at the point $(1, 2)$ where $\mathbf{u} = (3, 4)$.

Qu. 3. Find the directional derivative of the function $f(x, y, z) = (y^2 + \sin z)e^{-x}$ at $(1, 2, \pi/2)$ in the direction that points towards $(1, 1, 1)$.

Qu. 4. Find the equations of the tangent planes at the given points:

a) $4z = x^2 + 4y^2$ at the point $(2, 1, 2)$;
b) $xy^2 + 3x - z^2 = 4$ at the point $(2, 1, -2)$;
c) $Ax^2 + By^2 + Cz^2 = K$ at the point (x_1, y_1, z_1);
d) $z = e^{xy^2}$ at the point $(1, 1, e)$.

Qu. 5. Evaluate $\displaystyle\iint_R (x - y)\, dA$ as a repeated integral where R is the triangle with vertices at $(3, 10)$, $(3, 1)$, $(-2, 1)$. [Hint: first find the equations of the lines of the triangle.]

Qu. 6. By evaluating an appropriate double integral, find the area enclosed between the two parabolas $x = y^2$ and $x = 2y - y^2$.

Qu. 7. Sketch the region $R = \{(x, y, z) : 0 \le z \le 1 - |x| - |y|\}$ and evaluate the integral
$$\iiint_R (xy + z^2) dV.$$